

Target validation paths — osteosarcoma-mets-dll3-h7r2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

DLL3 IHC (clone SP347) is the test the case hinges on. A positive result at the trial threshold opens the only within-scope DLL3-directed therapy on the page (tarlatamab via NCT06788938). A negative result forecloses the entire DLL3-directed pathway — every BiTE, ADC, radioligand, and cell-therapy program in the dossier gates on protein-level confirmation, not transcript expression.

DLL3 RNA expression

Before any therapy decision: DLL3 IHC SP347 on archival FFPE, $\geq 1\%$ (preferably $\geq 25\%$) per NCT06788938's enrollment threshold. Turnaround is one to three weeks; cost is trivial relative to a treatment cycle. The first practical step is confirming SP347 assay availability at the treating institution — not every reference lab carries the Roche Tissue Diagnostics clone used in the tarlatamab development program.

Two refinements sit one tier down. Spatial heterogeneity is a known confounder in solid-tumor DLL3 (Zhang 2023): if multiple tumor blocks are available, IHC on a metastatic site as well as the primary refines confidence in whether the gating result generalizes. Neuroendocrine context (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1 panel) is research-grade for an osteosarcoma — DLL3 is normally a Notch-pathway target on neuroendocrine lineage, and an unexpectedly positive DLL3 IHC in a non-NEC tumor is worth contextualizing before the trial enrollment paperwork.

Germline TP53 sequencing (Li-Fraumeni panel) is a separate kind of finding — it does not affect tarlatamab eligibility, but late-teens / twenties osteosarcoma carries a meaningful prior probability for germline TP53. A positive result reshapes radiation-sensitivity considerations, screening for synchronous tumors, and family screening. Discuss with a genetic counselor before testing.

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.